

CHAPTER 4: NON-FUNCTIONAL REQUIREMENTS

4.1 PERFORMANCE

Performance requirements define the speed, throughput, and responsiveness standards the A9 Express platform must meet. Given that the target user base includes working professionals (30%), university students (25%), and families (20%) who rely on fast and dependable booking, performance is a critical quality attribute.

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-P01	Homepage and search results performance	Homepage and search results load within 2 seconds on a standard 4G/broadband connection.	High	Client (A9 Express)	Proposed
NFR-P02	Seat selection and booking confirmation speed	Pages respond within 3 seconds when 500 concurrent users are active.	High	Patient	Proposed
NFR-P03	High concurrency support	System sustains 500 concurrent users with no more than 5% error rate during load tests.	High	Admin	Proposed
NFR-P04	Airport hire and tour search speed	Search results returned within 3 seconds for any query.	Medium	System Design Team	Proposed
NFR-P05	Payment processing performance	Transactions complete within 5 seconds under normal load; users notified of failure within 10 seconds.	High	Dev Team	Proposed
NFR-P06	Mobile app data efficiency	Active use consumes no more than 50 MB of mobile data per hour.	Medium	Mobile UX Best Practices	Proposed
NFR-P07	Dashboard and analytics performance	Dashboard loads within 4 seconds for datasets up to 10,000 booking records.	Medium	Admin	Proposed

4.2 RELIABILITY

Reliability requirements ensure that the platform operates continuously and correctly, even in the face of failures. Bus travel, airport transfers, and tour bookings are time-sensitive; any downtime directly impacts passenger travel plans and operator revenue.

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-R01	System availability	Measured uptime $\geq 99.5\%$ per calendar month, verified via uptime monitoring tools.	High	Client (A9 Express)	Proposed
NFR-R02	Server failure handling	Automated failover activates within 60 seconds of any single node failure with no data loss.	High	Cloud Infrastructure (AWS)	Proposed
NFR-R03	Backup management	Automated daily backups completed successfully; backups retained for 30 days and restorable within 2 hours.	High	System Design Team	Proposed
NFR-R04	Disaster recovery	RTO ≤ 2 hours; RPO ≤ 1 hour of data loss for any unplanned outage.	High	System Design Team	Proposed
NFR-R05	Failed payment recovery	100% of failed payments are rolled back and users notified within 30 seconds.	High	Stripe/PayPal Integration Spec	Proposed
NFR-R06	Error monitoring and alerts	Critical errors trigger automated alerts to the dev team within 2 minutes via monitoring service.	Medium	System Design Team	Proposed
NFR-R07	Maintenance scheduling	All scheduled maintenance occurs between 12 AM – 4 AM with 24-hour advance notification to users.	Medium	Operational Policy	Proposed

4.3 USABILITY

Usability requirements ensure that all user groups — passengers, operators, and agents — can interact with the platform effectively and efficiently. Survey data (118 respondents) indicates that user adoption resistance is a key risk; therefore, simplicity and accessibility are prioritised.

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-U01	Easy booking for new users	First-time users complete booking in under 5 minutes in usability testing with no assistance.	High	User Survey (118 responses)	Proposed
NFR-U02	Multi-language support	Sinhala, Tamil, and English are fully supported; language switchable from any screen.	High	Client (A9 Express) / Target Market	Proposed
NFR-U03	Accessibility support	Passes WCAG 2.1 Level AA automated and manual audit with zero critical violations.	Medium	System Design Team	Proposed
NFR-U04	Timely user notifications	SMS and in-app notifications delivered within 60 seconds of booking confirmation, cancellation, or schedule change.	High	Client Requirement	Proposed
NFR-U05	Easy operator management	Operators complete fleet, route, schedule, and pricing setup within 10 minutes during onboarding testing.	High	Client (A9 Express)	Proposed
NFR-U06	Clear error messages	All error messages pass user comprehension testing: $\geq 90\%$ of test users understand the error and next step.	Medium	System Design Team	Proposed
NFR-U07	Hybrid booking support	Phone-assisted booking option available; confirmed bookings processed within the same system in under 5 minutes.	Medium	Target Market Analysis	Proposed

4.4 SECURITY

Security requirements protect user data, payment information, and platform integrity. Survey feedback identified payment security (68.6%) and reliability (53.4%) as the top concerns among potential users, making security a high-priority dimension across the system.

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-S01	Secure data transmission	All endpoints use HTTPS/TLS 1.2+; HTTP requests are redirected to HTTPS. Verified via SSL Labs scan score $\geq A$.	High	System Design Team / OWASP	Proposed
NFR-S02	Secure authentication	Clerk-managed authentication with OTP/MFA enabled; zero authentication bypass incidents in penetration testing.	High	Technology Stack (Clerk)	Proposed
NFR-S03	Secure payment processing	PCI-DSS compliant payment flow via Stripe/PayPal; no raw card data stored on platform servers — confirmed by security audit.	High	Stripe/PayPal Integration Spec	Proposed
NFR-S04	Secure payment processing	RBAC enforced: passengers, operators, agents, and admins cannot access unauthorised resources — validated by role-based test suite.	High	System Design Team	Proposed
NFR-S05	Fraud detection	Fraud detection flags $\geq 95\%$ of simulated fraudulent transactions in testing; blocked accounts notified within 30 seconds.	High	Risk Analysis	Proposed

NFR-S06	Secure password storage	All passwords hashed with bcrypt (cost factor ≥ 12); verified by code review — plaintext passwords absent from all storage layers.	High	Security Best Practices	<i>Proposed</i>
NFR-S07	Agent verification	100% of agent accounts require document upload and admin approval; unverified agents cannot process bookings.	High	Client Requirement / Risk Analysis	<i>Proposed</i>
NFR-S08	Session timeout	Sessions automatically expire after 30 minutes of inactivity on all platforms; user prompted to re-authenticate.	Medium	System Design Team	<i>Proposed</i>

4.5 SCALABILITY

Scalability requirements ensure the platform can accommodate growth in users, operators, and service types without requiring a full system redesign. The platform is expected to onboard multiple transport operators beyond A9 Express as the business grows.

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-SC01	Independent service scaling	Each service (booking, payment, notifications) scales independently; verified by load test showing no cascading failures.	High	System Design Team	<i>Proposed</i>
NFR-SC02	Automatic scaling during traffic spikes	AWS Auto Scaling provisions additional capacity within 3 minutes of CPU/load threshold breach with no downtime.	High	Cloud Infrastructure (AWS)	<i>Proposed</i>
NFR-SC03	Large database handling	Query response time remains under 2 seconds with 1 million booking records in the database.	High	System Design Team	<i>Proposed</i>
NFR-SC04	Easy onboarding of new operators	A new transport operator or service type can be onboarded within 2 developer-days without core architecture changes.	Medium	Client (A9 Express) Growth Plan	<i>Proposed</i>
NFR-SC05	Data caching efficiency	Cache hit rate $\geq 70\%$ for routes, schedules, and seat availability queries — measured via Redis monitoring.	Medium	System Design Team	<i>Proposed</i>
NFR-SC06	High API request handling	API sustains $\geq 1,000$ requests/minute per operator with $< 1\%$ error rate under load testing.	Medium	System Design Team	<i>Proposed</i>

4.6 MAINTAINABILITY

Maintainability requirements ensure that the system can be efficiently updated, debugged, and extended over its operational lifetime. The team has a 4-month development timeline followed by ongoing maintenance; a clean, documented codebase is essential for long-term project health.

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-M01	Consistent and documented codebase	ESLint/Prettier pass with zero errors on CI; $\geq 80\%$ of functions have inline documentation comments.	High	System Design Team	Proposed
NFR-M02	Automated testing and deployment	CI/CD pipeline (GitHub Actions) executes all tests and deploys to staging on every pull request merge with zero manual steps.	High	Technology Stack (GitHub)	Proposed
NFR-M03	Automated test coverage	Unit and integration test coverage $\geq 70\%$ across booking, payment, and authentication modules — measured by Jest coverage report.	Medium	System Design Team	Proposed
NFR-M04	API documentation	OpenAPI/Swagger documentation published and up to date; new developers complete onboarding within 5 working days.	Medium	System Design Team	Proposed
NFR-M05	Logging and audit support	Structured JSON logs retained for ≥ 90 days; log search returns results for any event within 10 seconds.	Medium	Operational Policy	Proposed
NFR-M06	Dependency maintenance	Dependency audit run quarterly; critical/high severity vulnerabilities patched within 14 days of disclosure.	Low	System Design Team	Proposed

4.7 PORTABILITY

Portability requirements ensure that the A9 Express platform can run across different devices, browsers, and deployment environments. Given that the target market spans mobile-first university students and working professionals using varied devices, broad device and browser compatibility is essential.

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-PO01	Cross-browser compatibility	Fully functional on Chrome, Firefox, Safari, and Edge (latest 2 versions); zero critical browser-specific bugs in QA.	High	System Design Team	Proposed
NFR-PO02	Cross-platform mobile support	Single React Native codebase deployed to Android (v8.0+) and iOS (v13+); all core features functional on both platforms.	High	Technology Stack	Proposed
NFR-PO03	Responsive web design	Layout renders correctly and all features are usable at screen widths from 360px to 1920px — verified by device emulation testing.	High	System Design Team / Target Market	Proposed
NFR-PO04	Cloud deployment portability	Services containerised with Docker; deployment to a different cloud provider (e.g., GCP) achievable within 1 working day.	Medium	System Design Team	Proposed
NFR-PO05	Database migration support	Schema documented and migration scripts prepared; estimated migration effort to PostgreSQL ≤ 2 weeks.	Low	System Design Team	Proposed
NFR-PO06	External configuration management	All secrets and environment variables stored in .env files / secret manager; zero hardcoded credentials in the codebase confirmed by code scan.	Medium	DevOps Best Practices	Proposed